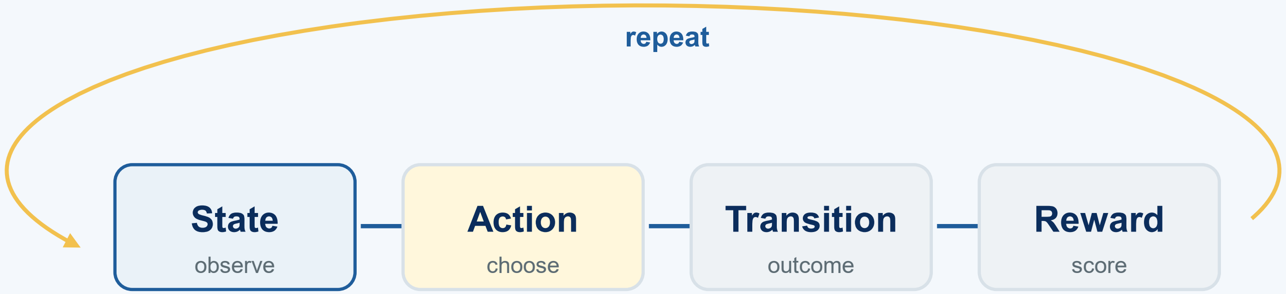
CompSci 471

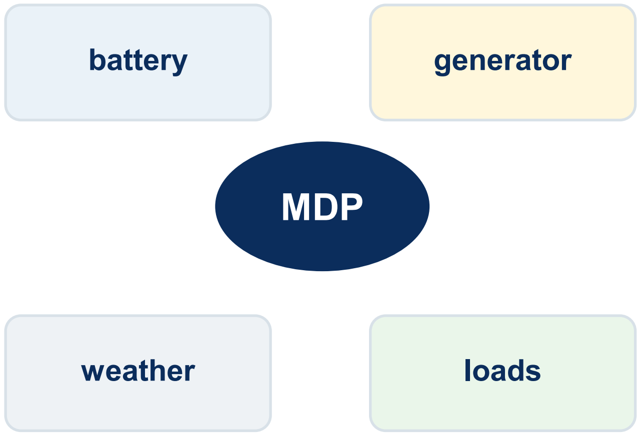
Artificial Intelligence

Lesson 11: Markov Decision Processes I

Observe → Choose → Transition → Reward → Repeat



Expeditionary microgrid controller



An MDP models repeated decisions by separating what the agent knows, chooses, observes, and values.

Fall 2026

Retrieval warm-up: what changed?

Module 1 planned through known models. Module 2 adds repeated decisions and uncertain outcomes.

Search problem

current state
available actions
goal test
path cost
solution path

Sequential decision problem

act now
observe outcome
receive reward
decide again
repeat

What if taking the same action does not always produce the same next state?

From search to MDPs

We are extending problem formulation, not replacing it.

Search	MDP
State	State
Action	Action
Successor state	Distribution over possible next states
Goal + path cost	Rewards over time
Solution path	Eventually: policy

Key shift: the agent chooses an action, but the environment may respond in several ways.

Module 2 map

MDPs are the common language for the solution methods that follow.

L11	L12	L13	L14	L15
Model the decision problem	Describe behavior and value	Value iteration	Policy iteration	Q-learning
S, A, transitions, rewards	policy, return, V, Q	Bellman backups	evaluate → improve	learn from experience

Today builds the environment model. Later lessons use that model to compute or learn better decisions.

Today’s mission

By the end of Lesson 11, students should be able to…

1. Define states, actions, transition models, and rewards
2. Explain the Markov property
3. Formulate real-world sequential problems as MDPs

Performance target

Given a new scenario, construct a reasonable MDP and explain whether its state contains enough information to predict what may happen next.

Running scenario: expeditionary microgrid

A controller must repeatedly manage power under changing conditions.

System

solar power
battery bank
backup generator
critical loads
changing demand

Possible actions

use battery
start generator
charge battery
shed noncritical load
maintain configuration

Uncertainty

cloud cover changes solar output
demand rises or falls
equipment may fail
battery performance varies

Question: what must the controller know before choosing an action?

Why a fixed plan is insufficient

A fixed plan assumes the world cooperates. An MDP supports adaptation after outcomes.

Solar
use available power

Charge
fill battery

Battery
power loads

Generator
backup

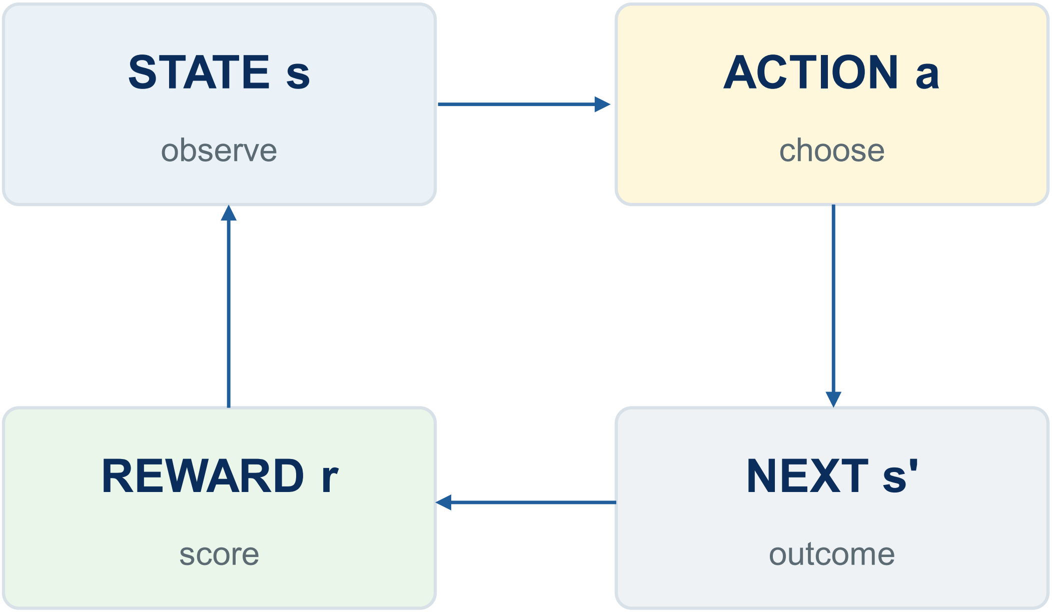
Then reality changes

clouds reduce solar
demand increases
generator start fails
battery charges slowly

The next decision should depend on what actually happened.

The MDP interaction loop

The agent chooses the action. The environment determines the outcome.



Agent controls

which action is selected from the current state

Environment responds

next state and reward according to the **transition model**

Observe → choose → experience outcome + reward → repeat

The four MDP components

An MDP separates the situation, the choice, the possible outcome, and the score.

State

What situation are we in now?

Action

What can the agent choose?

Transition model

What might happen next?

Reward

How desirable was the immediate outcome?

We will add policies, return, and value functions next lesson.

State = decision-relevant snapshot

A state should contain information needed for the next decision and prediction.

Microgrid state

(battery level,
generator status,
current demand,
solar condition,
critical-load status)

A good state helps determine

available actions
possible next states
transition probabilities
immediate rewards
the next decision

Connection to Lesson 4: states are still abstractions—but now they must also support prediction.

State representation still matters

For an MDP, a good state must support prediction, not just action selection.

Too little

generator status only

Cannot predict demand, battery safety, or start reliability.

Useful abstraction

battery + generator + demand + solar

Supports actions, outcomes, and rewards.

Too much

every sensor reading and serial number

May add complexity without improving decisions.

Markov preview: can two situations represented as the same state behave differently after the same action?

Actions are choices, not conditions

An action is something the agent can select from a state.

Action

Start generator
Shed noncritical load
Charge battery
Maintain configuration

Not an action

Battery at 20%
30% probability of failure
+10 reward
Generator starts successfully

Ask: is this under the agent's control as a choice?

Probability checkpoint: one action, several outcomes

Actions describe what the agent controls. Probabilities describe uncertainty in what happens next.

Action

Start Generator

Possible outcome	Probability
Starts normally	0.90
Remains off	0.08
Fault occurs	0.02

$$0.90 + 0.08 + 0.02 = 1$$

Ask

What did the agent choose?
What did the environment determine?

Reading transition notation

$P(s' \mid s, a)$ means probability of a next state given current state and action.

$$P(s' \mid s, a)$$

s	a	s'
current state	chosen action	next state

The vertical bar means "given"

We will study conditional probability formally in Lesson 18. Today, read what the statement says about this MDP.

In words: "How likely is this next state if we are here and take this action?"

Sixty-second probability check

Interpret transition probabilities without doing a full probability lesson.

Action

Move Forward

Outcome	Probability
Forward	0.75
Stays in place	0.20
Slips left	0.05

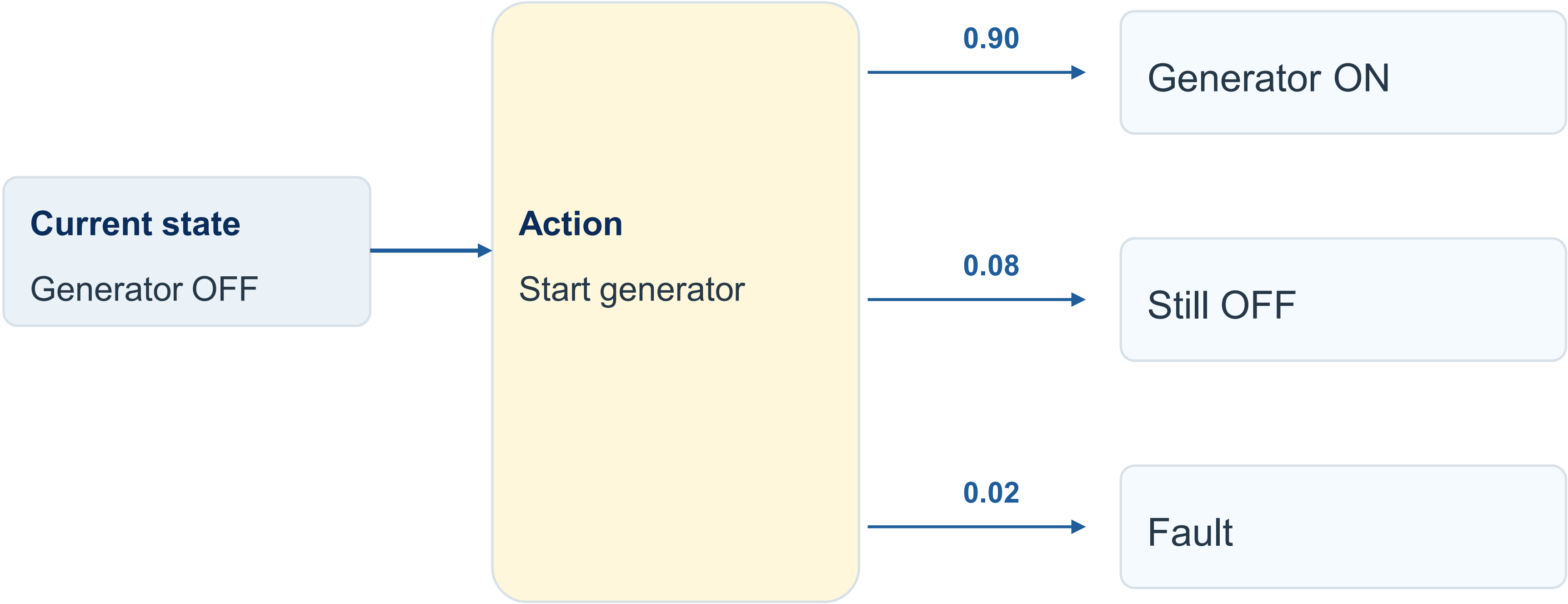
Questions

What was chosen?
Name one next state.
Valid probabilities?
Deterministic or stochastic?

No expected values yet. Today, probability only helps us read uncertain transitions.

Transition model = outcomes + probabilities

The transition model describes how the environment may move to a next state after an action.



Key checks

possible next states
probabilities
conditioned on s and a
sum to 1

Deterministic is a special case

An MDP can include deterministic actions, stochastic actions, or both.

Deterministic transition

Battery full + Use Battery
→ Battery medium

$P(s' \mid s, a) = 1$ for one outcome

Stochastic transition

Start Generator
→ starts, stays off, or faults

Several outcomes have nonzero probability

Stochastic simply means uncertainty remains after the action is chosen.

Action versus transition model

One is the choice. The other is the environment's response model.

Action	Transition model
what the agent chooses	what the environment may do
Start generator	90% on, 8% off, 2% fault
one selected choice	distribution of possible outcomes
under agent control	not completely under agent control

Diagnostic question: did the agent choose it, or did it happen after the choice?

Reward = immediate consequence

Reward is the immediate numerical signal associated with an outcome or transition.

Event	Reward
Critical systems remain powered	+10
Fuel consumed	−2
Unsafe battery level	−6
Critical outage	−20

Reward is not yet “how good the entire future is.” That comes next lesson.

Transition tells what happened; reward tells how we score it

The same action can lead to different outcomes and different immediate rewards.

Action

Use Battery

70% outcome

critical load maintained
reward +8

30% outcome

battery reaches unsafe level
reward −5

Check

Which numbers are probabilities?

Which numbers are rewards?

Transition = what happened. Reward = how desirable it was.

The agent optimizes what we reward

A mathematically valid reward can still encode the wrong objective.

Flawed reward design

+2 every time the controller switches power sources

+10 for maintaining mission power

→

Likely problem

The controller may switch repeatedly just to collect reward, even when switching is unnecessary.

The agent optimizes the reward it is given—not the designer's unstated intention.

Assemble the microgrid MDP

The four components define the decision environment that later lessons solve.

Component	Example
State	battery, generator status, demand, solar condition, critical-load status
Actions	use battery, start generator, charge, shed load, maintain
Transitions	probability distribution over resulting system states
Rewards	mission power, fuel use, safety, outage penalties

This is the problem model. L12–L15 will describe behavior, value, and learning inside this model.

The Markov question

Does the current state tell us enough to predict what happens next?

State = “generator is OFF”

Situation A

plenty of fuel
cool generator
low current demand

Situation B

nearly empty fuel
overheated generator
high current demand

Same recorded state. Same action. Different likely outcomes and rewards.

The Markov property

Relevant history should be summarized into the current state.

Student-facing definition

Once the current state and action are known, additional history should not be needed to predict the next-state distribution and immediate reward.

$$P(s_{t+1}, r_{t+1} \mid s_t, a_t, \text{history}) = P(s_{t+1}, r_{t+1} \mid s_t, a_t)$$

The current state must carry the relevant effects of the past.

Markov does not mean “history never matters”

Past events may matter because they changed current state variables.

History may matter

Prior generator use changed:
remaining fuel
equipment temperature
battery condition

But store the effects

State includes:
fuel level
temperature
battery level
current demand

We do not need every previous command if the current state summarizes what still matters.

The Markov test

Use these questions to evaluate a proposed state representation.

1. What actions are available?
2. After an action, what next states are possible?
3. Can we assign the right transition probabilities?
4. Can we determine the immediate reward?
5. Would additional history change those answers?

If #5 is yes, the state may be incomplete.

Debug the MDP

Find category errors before trying to solve the decision problem.

Scenario

A farming robot repeatedly chooses whether to water, inspect, or wait. Soil moisture and forecast conditions affect crop response.

Flawed model

State: current field
Action: current weather
Transition: reward +5
Reward: crop becomes healthier

Your task

Repair the categories.

Then decide whether “current field” is Markov.

Formulation sprint

Build an MDP for a new sequential decision problem.

Scenario

A hospital support robot repeatedly transports supplies among departments.
Elevator delays, battery charge, urgent requests, and corridor congestion affect outcomes.

State

reasonable representation

Actions

two choices

Transition

one action with two outcomes

Probabilities

simple values such as 0.8 / 0.2

Reward

one positive and one negative

Markov

one-sentence justification

Synthesis + exit ticket

Identify S, A, T, R, then test whether the state is Markov.

Scenario

A coastal monitoring buoy controller decides whether to transmit data, conserve power, or perform diagnostics.
Battery charge, weather, and sensor status affect the result.

Component	Your answer
S	
A	
T	
R	

Markov question

What information might need to be added so the current state can predict next outcomes and immediate reward?

Next class: what should the agent do, and how good is that behavior over time?